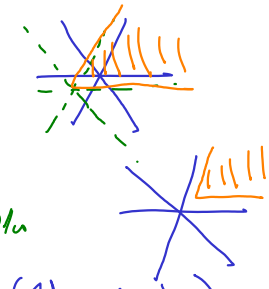
Proof of Beilinson-Bernstein



Recall: ① Γ is exact if $\langle \lambda + \rho, \alpha^\vee \rangle \notin \mathbb{Z}^{\leq -1}$

today → ② Γ is conservative if $\langle \lambda + \rho, \alpha^\vee \rangle \notin \mathbb{Z}^{\leq 0}$

$\Longrightarrow$ if both are satisfied, Bern-Beil calc. of $\Gamma(D_\lambda)$

$B \xhookrightarrow{?} G/U$

$$\Gamma : DMod(G/B, \lambda) \longrightarrow Mod(U\mathfrak{g} \otimes_{Z\mathfrak{g}} k_\lambda)$$

$G/U \to G/B$ torsor

$(\pi_* D_{G/U})^H \otimes_{k[\mathfrak{g}]} k_\lambda$ "strict monodromy"

"strict central character" i.e. $z - \chi(z) = 0$ $(\partial_\lambda \quad (z - \chi(z))^n = 0)$

Remark: non-strict variants are immediate at regular weights

Prop. Γ is conservative. (under above assumptions)

Proof:) antidominant Borel-Weil-Bott: $L_{w_0\lambda} \xrightarrow{\simeq} p_* \mathcal{L}_\lambda$

where: • $L_{w_0\lambda}$ simple G-rep with lowest weight λ

$\cong$ " " " highest weight $w_0\lambda$

naive dual → $(L_{-\lambda})^*$

not contragredient

• $p : G/B \longrightarrow$ pt

• $\mathcal{L}_\lambda$ = G-eq. line bundle on G/B $\longleftrightarrow$ B-reps : $\mathbb{C}_\lambda$

Define ⊛ $V_\lambda = p^* L_{w_0\lambda} = p^* p_* \mathcal{L}_\lambda \xrightarrow[G\text{-eq on } G/B]{} \mathcal{L}_\lambda$ counit of adj.

$\mathrm{Rep}(B) \ni \mathrm{Res}^G_B L_{w_0\lambda} \twoheadrightarrow k_\lambda$ filtration by weights, • sub highest weight, quot lowest weight

get a filtration of V_λ

Let $\mathcal{M}$ to be $(\pi_* D_{G/U})^H$-module w/ λ-monodromy.

& it's coherent over this ring, then

choose simple line bundle $\mathcal{L}$ such that

$\mathcal{M} \otimes \mathcal{L}$ is globally generated (as a $(\pi_* D_{G/U})^H$-module)

$\mathcal{L} = \mathcal{L}_\mu$ for μ very antidominant.

Consider: $\mathcal{M} \otimes V_\mu \twoheadrightarrow \mathcal{M} \otimes \mathcal{L}_\mu$ ← globally generated

Claim: map is split.

Why is this enough? $\Gamma(G/B, \mathcal{M} \otimes V_\mu) = p_*(\mathcal{M} \otimes p^* L)$

$\underset{\text{projection}}{=} p_*\mathcal{M} \otimes L = \Gamma(\mathcal{M}) \otimes L$

$\cup\!\!\mid\!\!\oplus$

$\Gamma(\mathcal{M} \otimes \mathcal{L}_\mu) \neq 0$.

If map splits, $\Gamma(\mathcal{M}) \neq 0$.

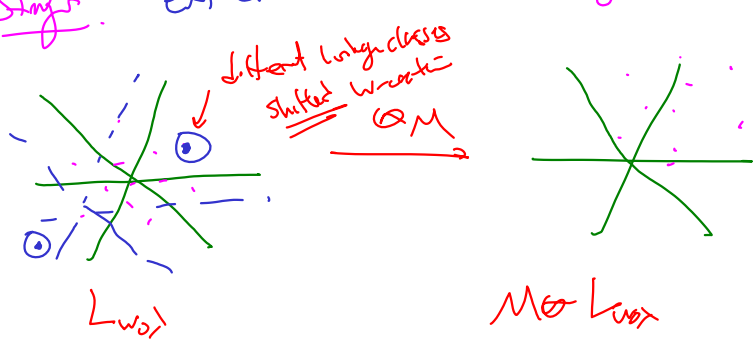
Proof of claim: $\mathcal{M} \otimes V_\mu$ has filtration, pieces: $\mathcal{M} \otimes \mathcal{L}_v$

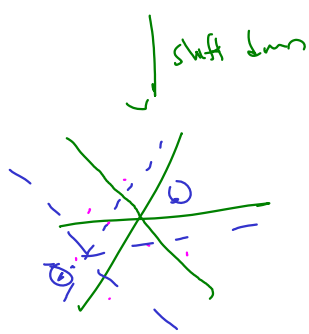
where v is a weight of $L_{w_0\lambda}$.

Know: $\mathcal{M}$ has λ-monodromy, $\mathcal{M} \otimes \mathcal{L}_v$ has $\lambda + v$ monodromy

Claim: $\lambda + v$ are in distinct linkage classes from $\lambda + \mu$ ← antidominant.

Strategy: extremal weights linkage classes [haven't used fact that $\lambda + \rho$ dom reg.]





Compatibility with translation functors

Def: $p: \tilde{X} \to X$ H-torsor for H torus

$\lambda, \mu \in \mathfrak{h}^*$ such that $\underbrace{\lambda - \mu}_{\chi} \in X^*(H) \subseteq \mathfrak{h}^*$ integral

$\mathcal{L}_\chi$ = sheaf of sections of line bundle on X: $\tilde{X} \times_H \chi$

$$\Theta_{\mu\lambda} := \mathcal{L}_\chi \otimes __ : DMod(\tilde{X}, \lambda) \to DMod(\tilde{X}, \mu)$$

monodromy

Exercise: This defines a functor on Derived Equivariant.

Prop: $\lambda + \rho$, $\mu + \rho$ dominant, $\lambda - \mu$ integral,

W-stabilizers of weights

$W^\lambda \subseteq W^\mu$

i.e. allowed to go deeper into the walls

$$\begin{tikzcd}
DMod(G/B, \lambda) \arrow[r, "\Gamma"] \arrow[d, "\Theta_{\mu\lambda}"] & Mod(U, \mathfrak{g}) \arrow[l, bend right, "Loc"'] \arrow[d, "T_{\mu\lambda}"] \\
DMod(G/B, \mu) \arrow[r, "\Gamma"] & Mod(U, \mathfrak{g}) \arrow[l, bend right, "Loc"']
\end{tikzcd}$$

Exercise ② $T_{\mu\lambda}$, $T_{\lambda\mu}$ biadjoint

Pf: $\nu = \lambda - \mu$ $\quad p: G/B \to pt$

Exercise ① There are pr_λ functors on both sides which commute w/ Γ, Loc

. Check that left adjoints commute

$$Loc(T_{\lambda\mu}(M))$$
$$= Loc(pr_\lambda(M \otimes L_\nu^*)) = pr_\lambda(Loc(M \otimes L_\nu^*))$$
$$= pr_\lambda(\underbrace{p^* M}_{\mu\text{-monodromy}} \otimes \mathcal{L}_{-w_0\nu})$$
$$= p^* M \otimes \mathcal{L}_{-\nu}$$

$-w_0\nu$ highest weight

$-\nu$ lowest weight

$\lambda = \mu + (-\nu)$

Claim: using filtration, the only λ-linked component is

What's left:
- figure out how to extract out $\mathcal{O}$
- what the vector were. $\rightsquigarrow$ Verma detly simple

Defn $G \curvearrowright X$ smooth.

- map of Lie algebra: $\alpha: \mathfrak{g} \to T(X) = \Gamma(X, T_X)$

$G^{(1)} = \mathrm{Spec}(k \oplus \mathfrak{g}^* \varepsilon) \simeq \mathrm{Spec}\, \mathcal{O}(G)/I_e^2$ $\quad$ I: ideal cutting out $e \in G$.

Coactin map (dual to $G \times X \to X$)

X not affine
$a: X \times G \to X$

$$a^* \mathcal{O}_X \xrightarrow{\text{act}^*} \mathcal{O}_{X \times G}$$

restrict to $G^{(1)}$

$$\mathcal{O}(U) \longrightarrow \mathcal{O}(U) \otimes \mathcal{O}(G^{(1)}) = \mathcal{O}(U) \oplus \mathcal{O}(U) \otimes \mathfrak{g}^* \varepsilon$$

① unit condition: $f \mapsto (f, \delta(f))$

② map of algebra: $(fg, \delta(fg)) = (f, \delta(f))(g, \delta(g)) = (fg, \delta(f) g + f \delta(g))$

$\therefore$ derivation

③ associativity: map of Lie algebra.

glue: $\delta: \mathcal{O}_X \to \mathcal{O}_X \otimes \mathfrak{g}^*$

To every $x \in X$ we have derivation δ_x

$\mathfrak{g} \to \mathrm{Der}(\mathcal{O}_X) = T(X)$

- G-eq. sheaf $\mathcal{M}$ $\quad$ "infinitesimal equivariance"

$\beta: \mathfrak{g} \to \mathrm{End}_{\mathcal{O}}(\mathcal{M})$ $\quad$ map of Lie alg.